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Unheard voices: the overlooked mental health toll of climate change in vulnerable communities

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SUMMARY

Climate change has an increasing physical and mental health toll on young people globally. In this Perspective, we suggest that the extent of mental health impacts is likely to be underestimated in the low- and middle-income countries which are most vulnerable to the effects of climate change. We highlight a strong global inverse relationship between internet connectivity and climate vulnerability, which poses significant challenges for understanding climate change's worldwide mental health impacts and for developing effective mitigation strategies. Inclusive methodologies that enable engagement with offline but climate-vulnerable communities are therefore needed. Such locally grounded mental health research is essential to ensure that climate policies are informed by the lived experiences of populations on the frontlines of the crisis, many of whom remain digitally disconnected and excluded from much current research.

CLIMATE CHANGE IS HAVING STRONG IMPACTS ON YOUTH MENTAL HEALTH IN LOW- AND MIDDLE-INCOME COUNTRIES

The accelerating impacts of climate change are reshaping societies across the world.[1] Because childhood and adolescence are developmentally sensitive phases of life, children and young people are uniquely susceptible to the consequences of climate change.[2] They are vulnerable to both its physiological effects and its psychological effects, such as climate-related worry and climate-related anxiety,[2–4] which in turn have adverse effects on concentration, performance and sleep.[2] While most of our knowledge of climate change's impact on youth mental health comes from work in high-income countries (HICs),[1] low- and middle-income countries (LMICs) are generally more vulnerable to climate change (figure 1a). This is because climate-related hazards interact with and destabilise interconnected systems, such as agriculture, healthcare, infrastructure and governance, and their impact is therefore most severe in already marginalised communities. Accordingly, climate change is known to have a pronounced impact on mental health in LMICs, including psychological distress, reduced well-being and emotional exhaustion, as reviewed by Marty and colleagues.[5]

The impact of climate change on youth mental health may be even stronger in those LMICs where climate change is already having clear and measurable effects on the environment. From 2018 to 2022, prolonged droughts in Southern Madagascar (the 'Grand Sud'), which faces a convergence of environmental and developmental challenges, led to severe food insecurity and malnutrition, culminating in what has been described as the world's first 'climate-induced famine.' In 2024, young people in the Grand Sud shared with us their views on climate change.[6] Participants completed, in person, a widely used and well-validated measure of climate-related psychological distress: the Climate Change Worry Scale (CCWS[7]), a 10-item self-report measure designed to assess the extent of worry individuals experience in relation to climate change. It focuses on proximal, personal concerns including statements such as 'Thoughts about climate change cause me to have worries about what the future may hold.' Participants answered questions using a 5-point Likert scale ranging from never (1) to always (5), and figure 1b shows the distribution of the CCWS (averaged across the 10 questions) among these participants. For comparison we obtained published measurements of the CCWS or Climate Change Anxiety Scale (CCAS),[2] where scores could be disaggregated for participants under the age of 24. The CCAS is another widely used tool that assesses emotional responses such as fear and helplessness, as well as the extent to which these feelings interfere with daily functioning; the CCAS also uses a 5-point Likert scale. Our preliminary findings from the Grand Sud show higher levels of climate worry than those typically reported in CCWS and CCAS samples.

EXCLUSION AND INCLUSION IN GLOBAL MENTAL HEALTH RESEARCH

The effects of climate change on youth mental health are likely to be mediated by multiple factors, with multiple time-courses.[8] Comparing across contexts and countries is, however, hampered by a lack of standardised constructs and underuse of validated and quantitative tools. Appropriate tools, including the CCWS and CCAS, could help better reveal the extent and global patterns of risk and resilience. They also permit consistent measurements well-suited to detect mental health responses to seasonal temperature shifts, extreme weather events, etc, and may therefore be able to guide (inter)national adaptation policies, which currently pay limited attention to mental health. Yet the feasibility of deploying such measures varies markedly across settings. In remote regions such as the Grand Sud, the lack of validated local instruments, limited infrastructure and connectivity, and the practical challenges of recruiting and

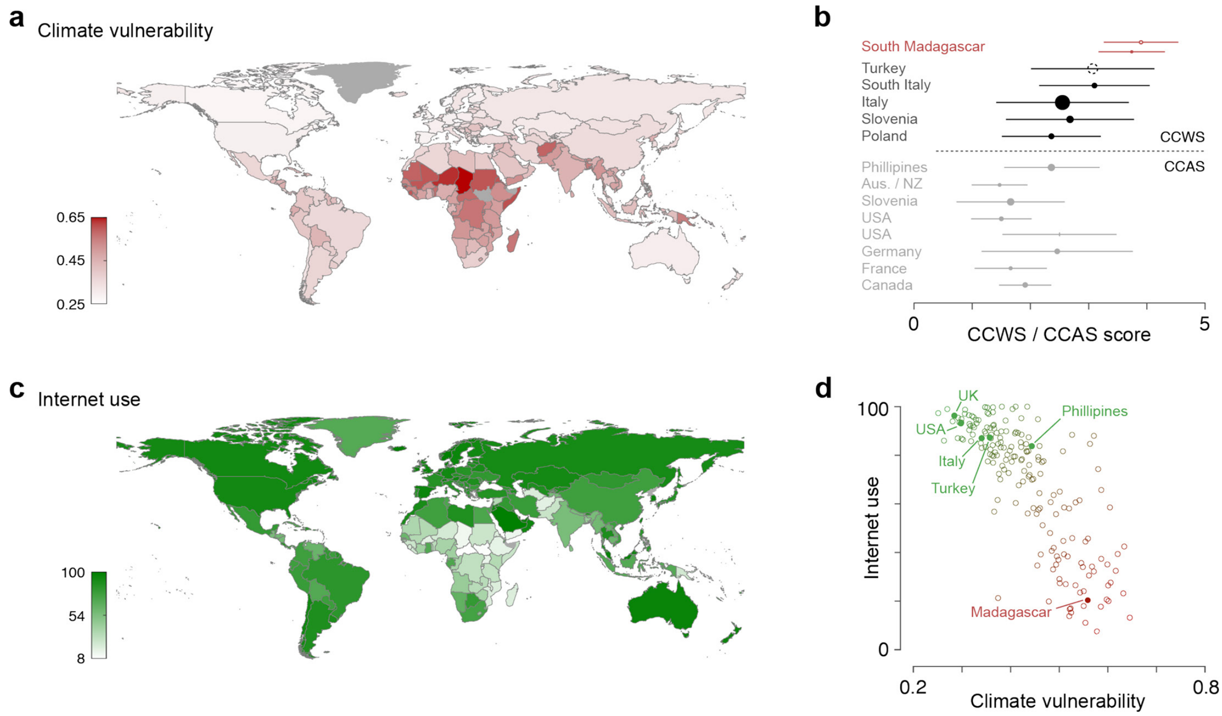

Figure 1 (a, d) Notre Dame Global Adaptation Initiative (ND-GAIN) 2024 (University of Notre Dame). Retrieved from https://gain.nd.edu/our-work/country-index/. (b) To address under-representation of adolescents and young adults in climate–mental health research, corresponding authors were contacted to obtain age-specific data. Unless otherwise specified, samples comprised participants aged 18–24 years. CCWS data were drawn from South Madagascar (n=23, ages 10–24),[6] Turkey (n=NR, ages 17–24),[3] South Italy (n=260),[10] Italy (n=1846, ages 13–21),[4] Slovenia (n=442)[18] and Poland (n=282).[11] CCAS data included the Philippines (n=425),[12] Australia and New Zealand (n=113),[13] the United States (n=23; n=185),[2 14] Germany (n=257),[15] France (n=131)[16] and Canada (n=247).[17] (c, d) International Telecommunication Union. Individuals using the internet (% of population) (dataset). Retrieved July 2025, from https://datahub.itu.int/data/?i=11624. CCAS, Climate Change Anxiety Scale; CCWS, Climate Change Worry Scale; NR, not reported.

assessing young participants make data collection difficult and necessitate adaptation to local conditions. These circumstances require methodological approaches that are context-responsive while remaining transparent about the constraints inherent in conducting research in marginalised communities.

Surveys and questionnaires are readily implemented via the internet and are therefore able to capture the views and worries of many participants. For example, a recent online survey across several HICs and LMICs was able to sample 10 000 youths and provide convincing evidence of high global worry about climate change.[9] The CCWS and CCAS have also been used to identify climate-related psychological distress in young people in different countries (figure 1b).[2–4 6 10–18] Notably, all but two of the studies using the CCWS collected data online.[3 6] However, as of early 2025, more than 2.6 billion people remain offline[19] and thus would not be included in internet-based studies. This reliance on online study implementation is particularly problematic because, as with climate vulnerability, internet access is unequally distributed (figure 1c). Internet usage is particularly low in many of the countries facing high climate stress—visual comparison of the maps in figure 1a,c shows a strong inverse relationship, and indeed the two measures are highly anti-correlated (figure 1d; $r(184)=-0.79$, $p<0.001$). A similar anti-correlation is also likely to be present within different regions of an LMIC—for example, none of our sample of Southern Madagascan youth (one of the most disadvantaged regions in Madagascar) had reliable electricity or access to the internet.

Finally, equity considerations also extend beyond data collection to the stewardship and use of the data. Research with marginalised and climate-vulnerable communities raises important questions about data ownership, how findings are shared and the extent to which results inform locally meaningful action. These issues are particularly salient where communities contribute data under challenging conditions yet face structural barriers to influencing policy or accessing the benefits of research. In the present study, findings were shared with local non-governmental partners and participant communities through non-technical written summaries and in-person discussions, refined in collaboration with local organisations to ensure cultural and contextual relevance. These feedback processes aimed to support local awareness and dialogue around climate-related mental health concerns and to ensure that study outputs were intelligible and useful to participants. Ensuring equitable data practices, including transparent governance, shared decision-making and attention to how evidence is actioned, will be essential as work in this area develops.

In summary, people in the world's most climate-vulnerable regions are the least able to participate in internet-based research or interventions. The severity of climate-related psychological distress may therefore be underestimated both within and across countries. Mixed-mode designs and offline recruitment approaches can mitigate this under-representation, but these strategies remain underutilised in large-scale quantitative studies using standardised mental health scales, particularly in highly climate-vulnerable settings.

FURTHER CONSIDERATIONS

It is important to note that our data derive from a preliminary mixed-methods study of limited sample size (n=83 for the quantitative survey; n=48 for focus groups). Although our climate-worry and climate-anxiety items were translated, back-translated and refined collaboratively with local partners, fully validated Malagasy-language instruments of climate change worry are not yet available, and the psychometric properties of these adapted measures remain to be established. Similarly, the regional sampling frame limits population-level inference. Within this context, the elevated levels of climate worry we observed should be interpreted as an early indication of potentially heightened climate-related distress in a highly climate-exposed context and of the need for further research using larger, representative samples and locally validated tools.

Furthermore, although the CCWS and CCAS can be placed on the same 1–5 scale, they tap into slightly different facets of climate-related distress: the CCWS emphasises the frequency and intensity of cognitive aspects of climate-related distress, whereas the CCAS incorporates affective and mild somatic anxiety symptoms in addition to cognitive processes. These distinctions mean that similar numerical scores may reflect different underlying response tendencies and symptoms. Furthermore, linguistic adaptation and cultural framing introduce subtle cross-language differences in how climate-related emotions such as 'worry' and 'anxiety' are conceptualised, leading to systematic measurement variation that explains why CCWS and CCAS show close but not identical alignment within the same samples.[18]

Alongside these practical considerations, there are also broader measurement challenges that shape how climate-related distress is studied globally. Recent evidence shows substantial inconsistency in how climate anxiety is defined and assessed in young people, with prevalence estimates varying markedly depending on whether validated scales or single-item questions are used.[20] In this landscape, the CCWS and CCAS are among the more developed instruments available, and we therefore used these as our reference points. Even so, further conceptual refinement and validation across diverse cultural and linguistic settings will be essential for producing truly comparable estimates and for distinguishing normative concern from clinically relevant distress. Finally, it is important to recognise that any online implementation is subject to selection bias. While this bias can be partially addressed through post-stratification or calibration weighting based on known population characteristics, online approaches should be viewed as complementary to, rather than a replacement for, carefully adapted in-person assessment.

CONCLUSION

This Perspective adds to those advocating an urgent need to better characterise, understand and mitigate the impacts of climate change on child and adolescent mental health in LMICs. Dominant epistemologies continue to shape what counts as evidence, often sidelining the experiential knowledge of communities, health workers and researchers in LMICs. The available evidence suggests that the mental health impact of climate change in LMICs is at least as great as in HICs, highlighting the need to include LMIC perspectives in the development of global strategies to mitigate these impacts. Our analyses illustrate a strong challenge to this inclusion, given by the global relationship between digital access and climate vulnerability. As mental health research and intervention move to exploit the capabilities of online platforms, climate-affected populations are at further risk of being excluded. Madagascar exemplifies this dilemma. Our in-person research in Southern Madagascar documented levels of climate-related psychological distress in adolescents that exceeded those reported in other countries; this underscores the importance of inclusive methods that reach beyond digital spaces, including working through schools, community centres or local health workers, using in-person or telephone approaches when connecting with participants who may have no or limited internet access, and ensuring offline methods of engagement remain available even when digital tools are used.

Contributors All authors contributed equally.

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